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Leveraging Local LLMs for Automated Changelog Generation

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SUSE

Outline

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Quickstart

Tumbleweed

```
#> sudo zypper in python-changelog-gen
#> cd SUPER_DUPER_PKG
#> osc-ai-vc
```

Leap 16.0

```
#> sudo zypper ar obs://science:machinelearning sc:ml
#> sudo zypper in python-changelog-gen
#> cd SUPER_DUPER_PKG
#> osc-ai-vc
```

Resources

https://huggingface.co/mslacken/t5-finetune-changelog
https://github.com/openSUSE/changelog-gen

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Introduction

Motivation

Changelogs in Open Build Service

- Package changelogs are created from `.changes` files
- Mostly reflect upstream changelog
- **But** must also contain changes for package creation

The Opportunity

- Accepted requests to `openSUSE:Factory` **are** the right format
- Submission requests contain the necessary information
- Perfect candidate for LLM automation

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Why use a Local LLM?

Existing Tools

- Existing tools can do the same
- Cannot capture changes to `.spec` or `_multibuild` files

Advantages

- **Hype Train**
- Pick up semantic changes (e.g., avoid Apple-specific changes)
- Runs completely locally, handling embargoed changes
- No information leak to cloud models

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Which Model to Choose?

Why Fine-Tune Instead of Training from Scratch?

- Pre-trained models understand language structure
- Training from scratch requires massive datasets (billions of tokens)
- Fine-tuning is computationally cheaper and faster
- Specializes existing knowledge for our specific task

Constraints

- Model must be small (around 1 GB)
- Permissive license
- Not quantized, so that it can be fine-tuned
- Transformer-based and not a GPT model

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T5 Model

Encoder-Decoder vs. Decoder-Only

Encoder-Decoder: T5, BART, mT5

- **Bidirectional encoder**: Sees entire input at once
- Best for: Translation, summarization, structured text generation

Decoder-Only: GPT-3, GPT-4, LLaMA, Mistral

- **Single decoder**: Input and output in same stream, but only sees previous tokens
- Best for: Text completion, general-purpose chat

Key Difference

T5 can **read all input first**, then generate. GPT processes input and output sequentially. This makes T5 more efficient for tasks with structured input.

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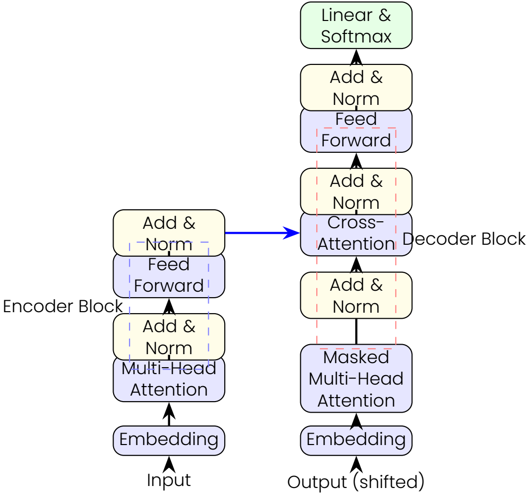
Encoder-Decoder Architecture (T5)

- **Encoder**: Bidirectional processing of the input sequence
- **Decoder**: Autoregressive generation with cross-attention to encoder



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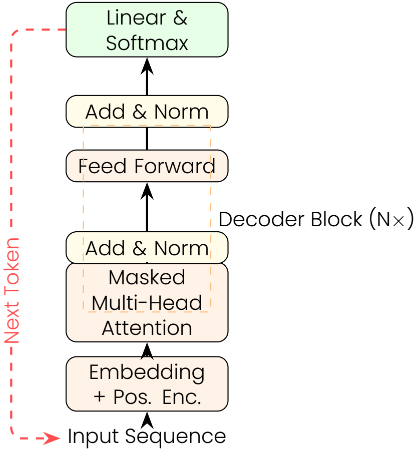
Decoder-Only Architecture (GPT/LLaMA)

- **Streamlined**: Unified architecture for both input and output
- **Causal Masking**: Each token only attends to previous tokens



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Attention Mechanisms: Full vs. Masked

Full Attention (T5 Encoder)

	create	changelog	for	pkg	gnome	music	44	45
create	0.14	0.08	0.19	0.12	0.14	0.17	0.1	0.06
changelog	0.19	0.15	0.09	0.07	0.07	0.19	0.08	0.16
for	0.15	0.17	0.14	0.09	0.12	0.19	0.07	0.07
pkg	0.13	0.1	0.15	0.11	0.13	0.16	0.15	0.07
gnome	0.09	0.06	0.09	0.06	0.23	0.29	0.09	0.11
music	0.05	0.16	0.11	0.18	0.13	0.12	0.2	0.04
44	0.18	0.12	0.07	0.14	0.2	0.15	0.09	0.05
45	0.13	0.2	0.06	0.2	0.16	0.07	0.08	0.1

All tokens considered

Masked Attention (GPT)

	create	changelog	for	pkg	gnome	music	44	45
create	1							
changelog	0.58	0.42						
for	0.36	0.22	0.42					
pkg	0.23	0.27	0.33	0.18				
gnome	0.2	0.18	0.21	0.2	0.2			
music	0.23	0.14	0.13	0.19	0.23	0.08		
44	0.05	0.2	0.07	0.15	0.16	0.2	0.17	
45	0.14	0.17	0.08	0.14	0.16	0.13	0.09	0.07

Only previous tokens considered

Numbers show attention weights (sum to 1.0 per row)
Darker green = stronger attention.

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T5: Origins and History

The T5 Paper (2019) by Google

- **T5 = Text-To-Text Transfer Transformer**
- Systematic study of NLP transfer learning
- **Everything is text-to-text**
- Apache 2.0 License

Pre-training

- Trained on C4 dataset (Colossal Clean Crawled Corpus)
- 750 GB of cleaned web text

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T5 Architecture: Encoder-Decoder

Encoder-Decoder Design

- **Encoder**: Processes input text
  - Bidirectional attention
  - Understands full context
  - Creates rich representations
- **Decoder**: Generates output text
  - Autoregressive generation
  - Attends to encoder outputs
  - Produces token-by-token

Model Sizes

Model	Params	Layers	Size
Small	60M	12	242 MB
Base	220M	24	892 MB
Large	770M	48	2.75 GB

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Data Collection Pipeline

OBS Data Extraction

What data do we need?

- Accepted requests to `openSUSE:Factory`
- Spec file diffs (version updates, build changes)
- Source file changes (added, removed, modified)
- Release notes from upstream (GitHub, archive changelogs)
- The actual `.changes` file entries (our target)

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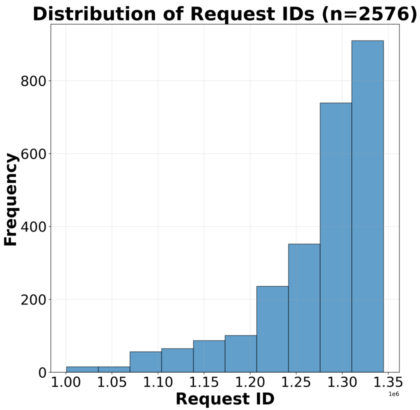
OBS Data Extraction

Naive Approach

- Choose a random number smaller than the latest request
- Check if submission went to `openSUSE:Factory` and was accepted
- Identify `Source0`
- Extract first 20 lines of `CHANGELOG` …
- For GitHub links, fetch GitHub release page



Use half-side Gaussian to get recent requests

OBS Data Extraction

Privacy Considerations

- Spec files use the MIT license; content can be freely used
- Changes files also use the MIT license
- `CHANGELOG` files fall under OSS licenses if accepted to `openSUSE:Factory`
- GitHub release pages use the same license as the source

Personal Information

- Submitter name and email are removed from changes files
- Reviewer information is not retrieved
- Request IDs are used as keys but not for training

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OBS Data Extraction - Example Update


```
{ "request_id": 1111873, "project": "GNOME:Factory", "package": "gnome-music", "version": "45.0",
   "changed_files": [ "gnome-music-45.0.tar.xz", "gnome-music.changes", "gnome-music.spec" ],
   "removed_files": [],
   "added_files": [],
   "spec_diff": "Name:           gnome-music\n-Version:        44.0\n+Version:        45.0\n Release:
    0\n Summary:        Music Player for GNOME\n License:        LGPL-2.1-or-later AND SUSE-GPL-2.0-with-
    plugin-exception\n Group:          Productivity/Multimedia/Sound/Players\n URL:            https://www.
    gnome.org\n-Source0:        https://download.gnome.org/sources/gnome-music/44/%{name}-%{version}.tar.xz\n
    +Source0:        https://download.gnome.org/sources/gnome-music/45/%{name}-%{version}.tar.xz\n #PATCH-FIX
    -SLE  0001-gnome-music-use-python36.patch yfjiang@suse.com -- disable python 3.7 specific feature to
    allow gnome-music build and run on python 3.6 for SLE/Leap 15.4.\n Patch0:         0001-gnome-music-use-
    python36.patch",
   "changes_diff": "+\n+- Update to version 45.0:\n+  + Minor quality of life improvements.\n+  + Updated
    translations.\n+\n - Update to version 44.0:",
   "source_file": "gnome-music-45.0.tar.xz",
   "archive_changelog": "Overview of changes in 45.0\n========================\n\•n Minor quality of life
    improvements\n\nThanks to our contributors this release:\n Carlos Garnacho\n Jean Felder\n Harold
    Knowlden\n Marinus Schraal\n\nUpdated or added translations:\n Abkhazian\n Romanian\n Italian\n Turkish\n
     British English\n French\n Hungarian\n German\n Hebrew\n Ukrainian",
   "url": "https://www.gnome.org",
   "github_release_notes": "" },
```


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OBS Data Extraction – Example Update Rendered

```
<INPUT>
create structured changelog for package gnome-music from 44.0 to 45.0:
changelog:
Overview of changes in 45.0
 ===========================
 • Minor quality of life improvements
 Thanks to our contributors this release:
 Carlos Garnacho
 Jean Felder
 Harold Knowlden
 Marinus Schraal
 Updated or added translations:
 Abkhazian
 [....]
 Hebrew
 Ukrainian
changes in spec file:
-Source0: https://download.gnome.org/sources/gnome-music/44/%<unk>name<unk>-%<unk>version<unk>.tar.xz
 +Source0: https://download.gnome.org/sources/gnome-music/45/%<unk>name<unk>-%<unk>version<unk>.tar.xz
</INPUT>
--------------------------------------------------------------------------------
<TARGET>
- Update to version 45.0:
  + Minor quality of life improvements.
  + Updated translations.
</TARGET>
```

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OBS Data Extraction - Example Patch File


```
{ "request_id": 1071207, "project": "network:utilities", "package": "pdsh", "version": "2.34",
  "changed_files": [ "pdsh.changes", "pdsh.spec" ],
  "removed_files": [],
  "added_files": [ "Add-call-to-slurm_init-this-makes-sure-the-config-options-are-set.patch" ],
  "spec_diff": "Patch6:         fail-fast-on-ssh-errors-or-non-zero-return-code.patch\n Patch7:         doc-
   fast-fail-update.patch\n Patch8:         Hack-to-work-around-a-generic-type-name-breakage-introduced-by-
   latest-Slurm.patch\n+Patch9:         Add-call-to-slurm_init-this-makes-sure-the-config-options-are-set.
   patch\n %description\n Pdsh is a multithreaded remote shell client which executes commands on\n %patch6 -
   p1\n %patch7 -p1\n %patch8 -p1\n+%patch9 -p1\n %build\n export CFLAGS=\"%{optflags} -fno-strict-aliasing\
   "",
  "changes_diff": "+\n+- Fix slurm plugin: make sure slurm_init() is called before using\n+  the Slurm API.
   This has been the case since version 20.11 (bsc#1209216).\n+  Add-call-to-slurm_init-this-makes-sure-the-
   config-options-are-set.patch\n+\n - Hack-to-work-around-a-generic-type-name-breakage-introduced-by-latest
   -Slurm.patch",
  "source_file": "pdsh-2.34.tar.gz",
  "archive_changelog": "This file describes changes in recent versions of pdsh. It primarily\ndocuments
   those changes that are of interest to users and admins.\n\n* Changes in pdsh-2.34 (2020-01-07)\n
   ===================================\n -- Fix for output corruption with no newlines (#114)\n -- pipecmd:
   fix result check error handling (Dylan Simon)\n -- slurm: workaround slurm export of internal List
   interfaces\n -- readline: add application name as argv[0] (#112)\n -- Fix errors from lgtm.com scan\n\n*
   Changes in pdsh-2.33 (2017-06-28)\n===================================\n -- Fix segfault and build issues
    on Mac OSX (#95)\n -- Always pass RTLD_GLOBAL to dlopen(3) of modules. Fixes missing symbol\n     errors
    from modules using libraries that also use dlopen() (e.g.\n     nodeupdown, slurm)\n\n* Changes in pdsh
   -2.32 (2017-06-22)\n===================================\n -- Autotools update\n -- Switch to dlopen(3)/
   dlsym(3) instead of using libltdl",
  "url": "http://pdsh.googlecode.com/",
  "github_release_notes": ""
},
```


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OBS Data Extraction - Example Patch File Rendered

```
<INPUT>
create structured changelog for package pdsh 2.34:
new files: ['Add-call-to-slurm_init-this-makes-sure-the-config-options-are-set.patch']
changes in spec file:
+Patch9: Add-call-to-slurm_init-this-makes-sure-the-config-options-are-set.patch
 +%patch9 -p1
</INPUT>
--------------------------------------------------------------------------------
<TARGET>
- Fix slurm plugin: make sure slurm_init() is called before using
  the Slurm API. This has been the case since version 20.11 (bsc#1209216).
  Add-call-to-slurm_init-this-makes-sure-the-config-options-are-set.patch
</TARGET>
```

Note

Upstream CHANGELOG missing on purpose, requests adds patch file

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What is Tokenization?

Tokenization

Converts text into numerical representations that models can process.

How It Works

- Text split into subword units (tokens)
- Each token mapped to unique ID
- Vocabulary built from training corpus
- Unknown words broken into known subwords

```
"Update to version 45.0"
```

↓ Tokenize

```
["Update", "to", "version", "45", ".", "0"]
```

↓ Map to IDs

```
[5974, 12, 988, 1902, 5, 209]
```

Subword example:

```
"changelog" → ["change", "log"]
"updating" → ["updat", "ing"]
```

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Data Preprocessing & Tokenization

Special Token Handling

- Newline character (`\n`) added as special token
- Preserves structure in multi-line changelogs

Sequence Length Limits

- **Input**: Maximum 1024 tokens (spec diffs + changelog)
- **Target**: Maximum 512 tokens (generated changelog entry)
- Sequences exceeding limits are truncated

Padding Strategy

- Dynamic padding: sequences padded to `longest` in batch
- Padding tokens in labels replaced with -100
- Loss function ignores -100 (no gradient from padding)

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Fine-Tuning & Results

Untrained Models: Fine-Tuning is Necessary

Request: 1111873

small

```
Minor quality of life improvements seitens unserer contributors this release: comprised of the following:
     likewise: embodied and added translations: characterized by a re-written version of the gnome-music from
     44.0 to 45.0: utilised changelog: assemble changelog for package gnomes-music between 44.0 and 45.0 :
     räte changelog - rätze changelog. rätsche -Source0: https://download.gnome.org/source
```

base

```
:========= .tar.xz  .tr.xs ăcprès +Source0: https://download.gnome.org/sources/gnome-music/Retrouvez +==:
     Marinus Schraal ==Hungarian==rätsel près +Retrouvez +Source1: httpswww.gno
```

large

```
-music from 44.0 to 45.0: ======= erweisen erweisend erweisendlich Peter Deutscherweise Germanerweise
     Germangegenseitig erweiselich Germanerweise Englishgegenseitigerweise Englisherweiseerweise
     Englishgebührenpflichtig Germanerweise Polisherweise Polisherweise -Source0:  --music ==-music-music
     erweise -erweise == =
```

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Untrained Models: Fine-Tuning is Necessary

Request: 1071207

small

```
-pdsh 2.34::-call-to-slurm_init-this-makes-sure-the-config-options-are-set.patch'] аеое о аи ереади в ил и
     рораии ирадиии орамителноо
```

base

```
structured changelog for package pdsh 2.34:îcreasing structured changelogging for package create structured
     change log for package build pcsh pkg 2.34.34:Auffinden Sie neue files: ['Add-call-to-slurm_init-this-
     makes-sure-the-config-options-are-set.patch']Währenden +%patch9 -p1: Add-call/slursm_install-this
```

large

```
 -p1
```

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Fine-Tuning Hyperparameters (Small Dataset)

Common Parameters

- **Optimizer**: AdaFactor
- **Learning rate scheduler**: Cosine
- **Evaluation Split**: 10%
- **Dataset**: ~3,200 samples

NVIDIA L40 GPU

- **Architecture**: Ada Lovelace
- **VRAM**: 48 GB GDDR6 (ECC)
- **CUDA Cores**: 18,176
- **TDP**: 300W

Model-Specific Parameters

Parameter	t5-small	t5-base	t5-large
Batch Size	30	10	3
Max Steps	21,000	64,000	12,468
Max Epochs	200	200	12

Note

Batch sizes scaled inversely with model size due to memory constraints

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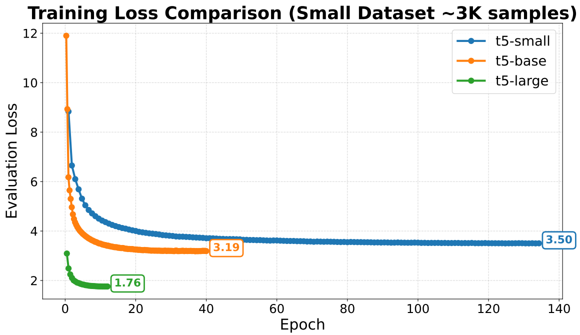
Training Results and Convergence (Small Dataset)

Training Loss Progression



Training Summary

Model	Steps	Runtime
t5-small	14,100	13h 08m
t5-base	12,800	14h 23m
t5-large	12,468	308h 56m

Stopping Reason

- **t5-small/base**: Early stopping (patience=5)
- **t5-large**: Max steps reached

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Fine-Tuned Model: First Results

Request 1111873 (gnome-music)

t5-small:

```
- Update to version 45.0:
 + Minor quality of life improvements.
 + Updated translations.
```

t5-base:

```
- Update to version 45.0:
 + Minor quality of life improvements.
```

t5-large:

```
- Update to version 45.0:
 + Minor quality of life improvements.
```

Request 1071207 (pdsh patch)

t5-small:

```
- Add add-call-to-slurm_init-this-makes-
sure-the-config-options-are-set.patch
```

t5-base:

```
- Add add-call-to-slurm_init-this-makes-
sure-the-config-options-are-set.patch
 * Fixes build with newer version of SLE15
```

t5-large:

```
- Add-call-to-slurm_init-this-makes-sure-
the-config-options-are-set.patch
```

Fine-Tuned Models: Results Discussion

Good

- No unrelated gibberish
- Mostly reflects the changes

Ugly

- **Overfitting**:

```
Updated translations
```

- **Hallucinations**:

```
Fixes build with newer version of SLE15
```

Overfitting

Small model summarizes a list of languages to "`Updated translations`"?

Hallucination

Input doesn't contain any SLE15:

```
create structured changelog for package pdsh 2.34:
new files: ['Add-call-to-slurm_init-this-makes-sure-the
    -config-options-are-set.patch']
changes in spec file:
+Patch9: Add-call-to-slurm_init-this-makes-sure-the-
    config-options-are-set.patch
 +%patch9 -p1
```

- Other samples show similar errors or are worse
- Bigger dataset needed

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Fine-Tuned Models: Scaling Up the Data

Problem

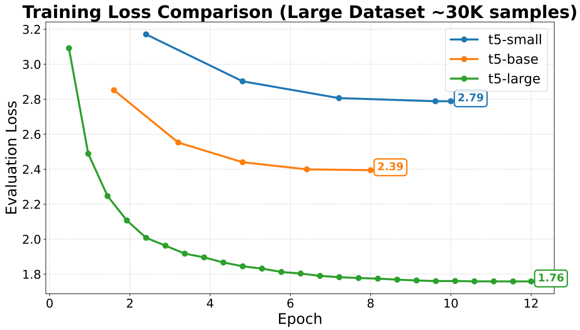
- Collection of dataset took two days
- Approximately 1 in 10 requests go to `openSUSE:Factory` and are accepted
- `osc request list -P openSUSE:Factory -s accepted` times out

Solution

- Adrian Schröter provided a list of **all** accepted requests
- Could easily create a list of 30,000 samples

Training Results and Convergence (Large Dataset)

Training Loss Progression



Training Summary

Model	Steps	Runtime
t5-small	2,080	1h 52m
t5-base	2,496	4h 56m
t5-large	12,468	23h 56m

Stopping Reason

- **t5-small**: Reached max epochs (10)
- **t5-base**: Reached max epochs (8)
- **t5-large**: Max steps reached (12)

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Fine-Tuned Models: Large Dataset Results

Request 1111873 (gnome-music)

t5-small:

```
- Update to version 45.0:
 * Minor quality of life improvements
 * Thanks to our contributors this release
```

t5-base:

```
- Update to version 45.0:
 + Minor quality of life improvements.
```

t5-large:

```
- Update to version 45.0:
 + Minor quality of life improvements.
```

Request 1071207 (pdsh patch)

t5-small:

```
- Add-call-to-slurm_init-this-makes-sure-
the-config-options-are-set.patch
```

t5-base:

```
- Add Add-call-to-slurm_init-this-makes
sure the configuration options are
set.patch
```

t5-large:

```
- Add-call-to-slurm_init-this-makes-sure-
the-config-options-are-set.patch
```

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Fine-Tuned Models: Large Dataset Discussion

Good

- No hallucinations
- Small model includes too much text as it lacks semantic understanding

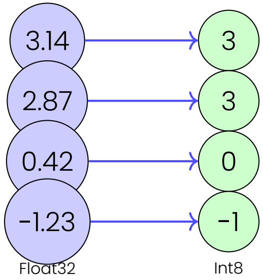
Bad

Base and **large** models are too large to run on laptops (> 1 GB)

Solution

Use quantization to reduce model size and runtime

Quantization



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Quantization & Deployment

What is Quantization?

Quantization

Reduces numerical precision of weights from high precision (Float32) to lower (Int8).

Why It Works

- Neural networks are **over-parametrized**
- Robust features tolerate small numerical errors
- Weight values cluster
- Activations follow predictable distributions

Benefits

- **Smaller size**: Float32 → Int8 = 4× reduction
- **Faster inference**: Integer operations are faster
- **Lower memory**: Fits on consumer hardware
- **Minimal accuracy loss**: Usually <2% degradation

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Quantization Implementation

CTranslate2 vs. Ollama

- **Ollama**: Optimized for decoder-only models (GPT, LLaMA)
- **CTranslate2**: Optimized for encoder-decoder models (T5, BART)
- T5's bidirectional encoder requires specialized attention mechanisms

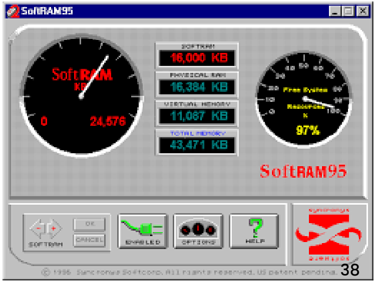
INT8 Quantization

- Float32 (4 bytes) → Int8 (1 byte)
- **Result**: ∼4× smaller model size
- **t5-base**: 892 MB → ∼220 MB
- **t5-large**: 2.75 GB → ∼690 MB

Compression Results

Model	Size
t5-small	60 MB
t5-base	220 MB
t5-large	690 MB

All models now fit in typical laptop RAM!



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Wrap-up

Model

- Model and quantized versions uploaded to Hugging Face
- `https://huggingface.co/mslacken/t5-finetune-changelog`
- FP32 model (full precision) for `ctranslate2`

Integration

- Simple Python integration available on GitHub
- `https://github.com/openSUSE/changelog-gen`
- `obs://science:machinelearning/python-changelog-gen`

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Future Plans – LabsConf 2027

Initial Python Package Generation

- Many python packages available
- **py2pack** struggles with multiflavor python builds
- More python packages sensible?

Cross distribution packaging

- Would be immensely useful, if working
- Too complicated to get it working?
- Works both ways!

Factory acceptance bot

- Check if package is fit for factory
- Package conflicts can't be caught
- Simpler answer, yay or nay

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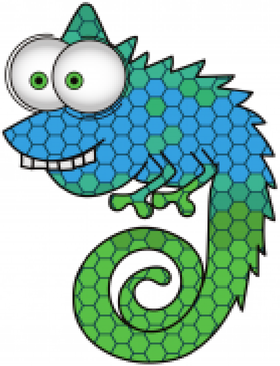
Thank you for your attention!



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Used Software & Availability

Core Training Packages

- `python-torch`
- `python-transformers` - Hugging Face T5
- `python-tokenizers` - Fast tokenization
- `python-safetensors` - Model serialization

Inference & Deployment

- `ctranslate2` - Quantization & fast inference

openSUSE:Factory Status

Package	Status
python-torch	✓
python-transformers	✓
python-tokenizers	✓
python-safetensors	✓
ctranslate2	✓

`python-changelog-gen`:
`science:machinelearning`

Note

All essential packages available in Factory but without acceleration!

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Disk Space Requirements

Total Space Used: **37 GB**

Component	**Size**
Virtual Environment (PyTorch)	5.7 GB
Training Code & Scripts	<100 MB
Datasets (small + large)	137 MB
Fine-tuned Models:	
Small dataset runs	13 GB
Large dataset runs	13 GB
Current workspace	5.6 GB

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